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Project Title:	Identification of the changes in the brain proteome associated with HZE-induced neurocognitive impairment		
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Key Personnel Changes/Previous PI:	November 2012 report: Dr. Sucharita Dutta, a proteomics expert, has been added to the team.		
COI Name (Institution):	Drake, Richard (Eastern Virginia Medical School) Nyalwidhe, Julius (Eastern Virginia Medical School) Semmes, Oliver (Eastern Virginia Medical School)		
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Task Description:	One of NASA's biggest concerns about the long-term health of astronauts who go on deep space missions is the impact that Hze radiation will have on brain function (neurocognition). Data from our laboratory and others suggests that there is significant impairment of certain neurocognitive tasks following exposure to low Hze doses. The goal of this application is to use proteomic profiling/imaging to establish the mechanistic basis of Hze-induced cognitive impairment. Our underlying hypothesis is that Hze-induced spatial memory impairments (HISMI) or Attentional Set Shifting Impairments (HIASSI) are due to Hze-induced changes in the proteome of the brain (neuroproteome). Our sub-hypothesis is that the use of a novel technique, called MALDI-MSI, will identify specific regions of the brain where the neuroproteome differs in rats that have developed HISMI and HIASSI. We further hypothesize that proteomic profiling of serum and cerebrospinal fluid (CSF) will identify proteins that can be used as biomarkers to monitor the emergence of HISMI and HIASSI. We also hypothesize that different mechanisms of HISMI and HIASSI occur following exposure to Hze radiations with different track structures. Our studies will thus address CNS Gap 2, and we shall specifically focus of the following aims: Aim 1. Identify changes in the neuroproteome that are associated with HISMI and HIASSI. Aim 2. Identify proteins that are present in the serum and CSF that are biomarkers of an individual rat¿s susceptibility to HISMI/HIASSI. Aim 3. Determine the LET-dependency for HISMI and HIASSI, and whether the threshold dose is modified by prior exposure to protons, and if the underlying mechanisms of HISMI and HIASSI is constant irrespective of the LET of the Hze particle. In Aim 1, two groups of rats (5-week and 9 month old males) will be irradiated with 20 cGy Hze (1 GeV 56Fe) irradiation, and HISMI and HIASSI assessed at 3 months post irradiation. The rats will then been classified as normal or as having HISMI or HIASSI, then euthanized and their brains recovered. We shall use MALDI-IMS to analyze changes in the neuroproteome at the regional level. Subsequently, we shall identify specific proteins whose expression changes when HISMI or HIASSI have emerged, and will use immunohistochemcial analysis to determine the specific brain tissue (e.g. the dentate gyrus or CA2) affected. These studies will give considerable insight into the underlying cause for Hze-induced neurocognitive failure. In Aim 2, biological fluids (CSF and serum) from the rats used in Aim 1 will be recovered, and then subjected to MALDI-TOF/TOF proteomic profiling to identify unique proteins that are prognostic markers of an individual's rat's susceptibility to develop HIMSI and HIASSI. These proteins could be potentially used for post-mission surveillance of astronauts to monitor the emergence of Hze-induced neurocognitive impairments. In Aim 3, we shall repeat the studies outlined in Aim 1, but determine the impact that exposure to other Hze particles (300 MeV 37Cl, 1 Gev 48Ti , 400 MeV 48Ti) have on the induction of HISMI and HIASSI, and determine whether the mechanism of HIASSI and HISMI is the same following exposure to these Hze radiations as was observed in Aim 1. These studies will be confined to the aged rat population, as these resemble the biological age of the current astronaut cadre. We shall also determine if prior irradiation with 1 Gy of protons modifies the threshold dose for HISMI and HIASSI and the spectrum of proteomic changes that occur when HIASSI and HISMI emerge. Due to logistical constraints these combined proton/Hze exposure studies will be confined to 400 MeV 48Ti and 1 GeV 56Fe. Collectively, these studies will provide considerable insight into the underlying mechanism of Hze-induced neurocognitive failure and potentially provide a post mission screen for HISMI and HIASSI in astronauts returning from deep space missions.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	Current models predict that the astronauts on deep space missions will be exposed to ~25 cGy of Galactic cosmic radiation (GCR). The long-term consequence of exposure to such doses is largely unknown, but recent data from our laboratory and that of Dr. Rabin, suggest that low (<20 cGy) doses of HZE particles lead to significant impairments of various neurocognitive functions. We have shown that exposure of young rats to 20 cGy of 1 GeV/u 56Fe leads to a significant impairment in spatial memory (Britten et al., 2012) and attentional set shifting performance (Lonart et al., 2012). 1 GeV/u 56Fe particles have a LET of 150 keV/n, that has been shown to be the "optimum" for cell killing. Data from Dr. Rabin suggest that the LET dependency for neurocognitive impairment may differ substantially from that seen for cell killing. We have thus started to determine the LET-dependency for impairment of Spatial memory performance and attentional set shifting in rats that have a comparable biological age to the astronaut population. Our first ion species comparison involved 1 GeV/u 56Fe and 1.1 GeV/u 48Ti, with LETs of 150 and 106 keV/um respectively. We have also started to investigate the mechanistic basis for the impairment of these various neurocognitive processes using an unbiased proteomic analysis. Due to the ease of extraction we have initially focused our efforts on determining the changes in the proteome of the hippocampus in rats that have good or poor spatial memory performance after exposure to these two ion species (1 GeV/u 56Fe and 1.1 GeV/u 48Ti). MATERIALS & METHODS. Retired breeder (6-9 months old) male Wistar rats were exposed to either 1 GeV 56Fe or 1.1 GeV 48Ti particles at BNL. At 3 months post exposure the ability of rats to conduct attentional set shifting and spatial memory was established. After completion of the behavioral tasks the animals were euthanized and the brains recovered for subsequent protoeomic analysis on the LTQ platform. RESULTS. Spatial memory impairment. Preliminary data from our first three cohorts of aged rats, suggests that Spatial memory impairment occurs following exposure to 10 cGy 1 GeV/u 56Fe. However, there is a marked inter-individual susceptibility to develop spatial memory impairment, with ~33% of rats exhibiting spatial memory performance comparable to that observed in unirradiated rats. Exposure of aged rats to 20 cGy of 1.1 GeV/u 48Ti ions resulted in an impairment of spatial memory performance, but in contrast to what was observed with 56Fe particles, no impairment was seen following exposure to 10 cGy 1.1 GeV/u 48Ti. Exposure to 15 cGy resulted in an intermediate response between what was observed with 10 and 20 cGy, suggesting that the threshold dose is ~15 cGy. Consideration of this collective data suggests there may be some LET dependency on the Threshold dose for spatial memory impairment, with 1 GeV/u 56Fe being more (>1.5 fold more) deleterious that 1.1 GeV/u 48Ti Attentional Set shifting impairment. We have recently published (Lonart et al., 2012) data from young rats that had received cranial irradiation with 1 GeV/µ 56Fe, where we reported that exposure to 20 cGy 1 GeV/µ 56Fe resulted in the majority of rats being incapable of completing all stages of the Attentional Set Shifting test, with many rats failing at the Reversal 1 stage. We repeated these studies using whole body exposures and observed a similar pattern of failures to that seen with the cranially irradiated rats. Sixty percent of the sham treated rats completed all stages of the testing whereas only 20% of the irradiated rats could complete all stages of the task. These rats have a biological age comparable to that of teenagers and our primary mandate is to determine if rats with a comparable age to the Astronauts were equally sensitive to the effects of these low doses of 1 GeV/u 56Fe and 1.1 GeV/u 48Ti. Our preliminary data from three cohorts of rats exposed to 15 & 20 cGy 1 GeV/µ 56Fe suggests that there is a significant impairment of attentional set shifting performance. However, there is an age related decline in the ability or the motivation of Aged rats to conduct Simple Discrimination that may mask the full impact of the HZE irradiation. Exposure to 15 & 20 cGy 1 GeV/µ 56Fe further decreases the ability of rats to perform Simple Discrimination, and there is also a decrease in the ability of rats to conduct Reversal 1 stage (as was observed in the juvenile rats). Interestingly exposure to 15 cGy 1 GeV/µ 56Fe, resulted in a greater impairment of ttentional set shifting capability than did 20 cGy, primarily due to the marked impairment of the ability of the rats to conduct complex discrimination (probably regulated by perirhinal cortical region). The impairment of Reversal 1 performance was also observed. This inverse dose effect appears to be a highly reproducible phenomenon being consistently observed in all three batches of rats studied. Exposure to 20 cGy 1.1 GeV/u 48Ti ions, also resulted in a significant impairment of the ability of aged rats to conduct Attentional set shifting. Approximately 20% of the sham-irradiated rats were unable to complete the SD stage, but further reduction in the ability of the rats to perform Simple Discrimination was observed following exposure to 20 cGy 1.1 GeV/u 48Ti ions. The irradiated rats were also significantly impaired in completely the Rev1 stage. In contrast to what was observed when 15 cGy with 1 GeV/u 56Fe was used, where there was a greater impairment of attentional set shifting ability, 15 cGy 1.1 MeV/u 48Ti resulted in less impairment of Simple and Complex Discrimination performance, but still resulted in a significant impairment of the ability to conduct the Rev1 stage. These data suggest that the basal forebrain and medial pre-frontal cortex may be especially sensitive to HZE (both 1 GeV/u 56Fe and 1.1 GeV/u 48Ti ions). Hippocampal proteomic changes. We have completed our first round of LTQ analysis on the pooled hippocampi from the various cohort of juvenile rats exposed to 20 cGy of 1 GeV/u 56Fe particles, and we have identified several changes (~16) in the neuroproteome that look promising as biomarkers of HZE-induced spatial memory impairment (HISMI), or resistance to HISMI. We have started to verify if these biomarkers proteins are truly changed within the hippocampus and started to investigate whether proteins whose activity is influenced by the biomarkers are also altered. These very preliminary data suggest that there are specific markers of spatial memory performance in these rats, however, to date we have not found any biomarker that is common to spatial memory impairment induced by both 1 GeV/u 56Fe and 1.1 GeV/u 48Ti ions. These data suggest that there may be different modes of HISMI induced by different HZE particles SUMMARY. These preliminary studies suggest that mission-relevant doses (10-15 cGy) of HZE particles with LETs of 106-150 KeV/u lead to significant neurocognitive defects (spatial memory performance and attentional set shifting) in rats of comparable biological age to the astronaut population. We shall continue to identify the threshold dose for this impairment and will determine if there is any LET dependency for that dose. Our preliminary data suggest that there may not be a universal LET dependency for impairment of all neurocognitive tasks. Executive function tasks primarily regulated by the medial prefrontal cortex and basal forebrain appear to be most sensitive to both 1 GeV/u 56Fe and 1.1 GeV/u 48Ti particles with roughly equal levels of impairment induced by each ion species. There may be some variation in the sensitivity of hippocampal dependent processed to ions of differing LET, with 1 GeV/u 56Fe leading to >1.5-fold greater inhibition than 1.1 GeV/u 48Ti particles. We have identified specific changes in the hippocampal proteome that are associated with either good or bad spatial memory performance in HZE-irradiated rats, that may provide some insight into the mechanisms that govern individual susceptibility to HZE-induced cognitive impairment.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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Abstracts for Journals and Proceedings	Britten RA, Davis L, Benzine T, Fellman L, Semmes J, Nyalwidhe J, Drake RR. "Changes in the hippocampal proteome associated with HZE-induced cognitive impairment." 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. Presentation Number: PS5-30. Abstract available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=f8c8e816-7bf5-4085-94f8-f076ca29978a&cKey=e4a57e04-7e81-4afd-8dcf-0cb4b50b1daf&mKey={1514526B-FDA6-42BC-948B-C58023F5A94E} ; accessed 11/20/2012. , Oct-2012